

ENGINE-SENSE AI — DIAGNOSTIC REPORT

Analysis ID: ANL-78188760 | Generated: 2026-09-20 19:55:48

OVERALL DIAGNOSIS: **ENGINE KNOCK / DETONATION**

Confidence Score: 80.0% | Severity Level: Critical

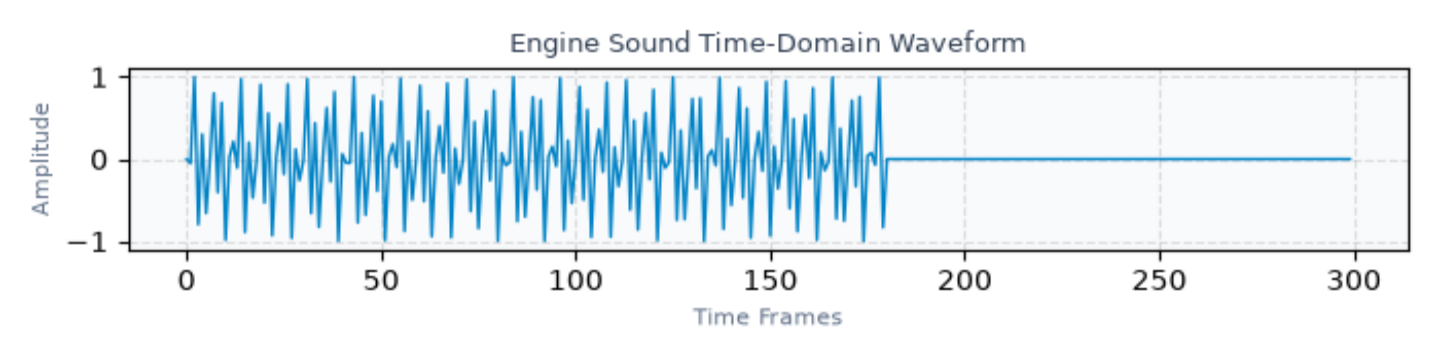
Vehicle & Recording Metadata

Vehicle Brand:	Toyota	Audio File:	sample_engine_normal.wav
Model / Engine:	Camry V6 (3.5L V6 DOHC)	Duration:	5.00 sec
Fuel / Mileage:	Gasoline / 52000 km	Processing Time:	6.210 sec

Multi-Model Predictions Breakdown

Model	Type	Predicted Class	Confidence
Random Forest Classifier	ML/DL Model (Untrained)	knocking	80.0%
Support Vector Machine (SVM)	ML/DL Model (Untrained)	knocking	80.0%
1D Convolutional Neural Network	ML/DL Model (Untrained)	knocking	80.0%
2D Mel-Spectrogram CNN	ML/DL Model (Untrained)	knocking	80.0%
LSTM Recurrent Neural Network	ML/DL Model (Untrained)	knocking	80.0%

Acoustic Waveform Analysis



Mechanical Recommendation

Check fuel octane rating, ignition timing, knock sensors, and carbon deposit buildup in cylinder head.

DISCLAIMER: This system provides an AI-assisted acoustic analysis and should not be treated as a substitute for professional mechanical inspection.